

Dynamic deep optics: learning controllable hybrid optical-digital systems.

Overview. Deep optics has emerged as a promising paradigm to push the limits of computational imaging systems, using machine learning methods to directly learn the best hybrid optical-digital system for a given task. Most existing methods assume static, non-changing optics and camera systems for snapshot captures. However, real camera systems are dynamic; they can rapidly change focus, zoom, adjust their aperture, or move through the world. The focus of this CAREER proposal is to bring deep optics to dynamic imaging systems; harnessing dynamic optics to improve information capture and computational requirements for hybrid optical-digital systems. Specifically, this proposal aims to:

1. Develop a toolbox for deep optics design for dynamic imaging systems. We will add differentiable focus, zoom, and aperture controls for end-to-end simulation of hybrid ray-wave optical systems.
2. Provide analysis of design trade-offs for dynamic imaging systems. How does optical design change under dynamic capture? We will address stability, robustness, and computational trade-offs.
3. Learn deep optics designs for several high-level tasks, then build and analyze prototype dynamic computational imaging systems fabricated using advanced grayscale lithography methods.

Intellectual Merit. This proposal addresses a key limitation and open question in the field of computational imaging: Are deep optical designs brittle under dynamic conditions, such as changes in focus and lens aperture? We will develop optical systems that not only generalize but also provide more information by leveraging dynamics as a degree of freedom during optimization. Rather than relying on monocular vision with strong, learned priors, we take an alternative approach: optimize the camera to capture the signal of interest (e.g., 3D cues, multiple wavelengths, polarization), then use lightweight, physics-informed digital compute to extract encoded signals from the sensor. By specifically optimizing lenses and optical phase masks for dynamic capture, we can harness more degrees of freedom for optical encoding. We expect that this will result in more powerful systems that can have better size, weight, and power trade-offs, and can be more computationally efficient by offloading power-hungry digital computation to passive advantageous optical encoding and computation.

The PI's prior work in physics-informed machine learning, deep optics for compressive 3D imaging, and dynamic multi-shot compressive hyperspectral imaging lays the necessary groundwork for this proposal. This proposal formalizes, combines, and extends these ideas to harness the potential of deep optics for dynamic computational imaging systems.

Broader Impacts. Advancing imaging technologies has important impacts on national security and economic competitiveness. Imaging systems are critical across multiple sectors; for example, they are used in advanced glass manufacturing for defect detection, and are critical for spatial understanding and navigation in robotics. New York State is a hub for advanced optics and photonics manufacturing and R&D. As part of this proposal, we will leverage and advance this excellence to increase workforce development and economic competitiveness. First, we will develop a Master's-level class at Cornell University on computational optics, with a substantial final project involving local manufacturing company partners, such as Corning. This will advance workforce development while bringing advanced computational imaging ideas to local companies. Next, we will organize a New York State computational optics workshop and recurring monthly seminar series to bring together researchers across academia and local industry to share recent advances and outstanding problems. Finally, we will engage with local rural high schools through imaging and signal-processing outreach activities to broaden participation in STEM and help develop the local workforce. These will further strengthen the economic competitiveness for advanced imaging and optics in New York State while also addressing workforce development.